

Does Remote Expert Representation really matters: A comparison of Video and AR-based Guidance

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ABSTRACT

This work describes a user study aimed at understanding how the remote expert representation affects the sense of social presence in scenarios of remote guidance. We compared a traditional video chat solution with an Augmented Reality (AR) annotation tool. These were selected due to ongoing research with partners from the industry sector, following the insights of a participatory design process. A well defined-problem was used, i.e., a synchronous maintenance task with 4 completion stages that required a remote expert using a computer to guide 26 on-site participants wielding a handheld device. The results of the study are described and discussed.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction Paradigms—Mixed/augmented reality; Human-centered computing—Human computer interaction (HCI)—HCI Design and Evaluation Methods—User Studies

1 INTRODUCTION

The sense of social presence goes beyond the simple information regarding the location or availability of a team member, as it entails a feeling of being present during the interaction and following what is happening [3]. Research hints that, as happens with co-located work, the sense of social presence might have an important influence on remote collaboration [1].

To support this type of scenarios, there has been a growing interest in using Augmented Reality (AR), a technology that enhances a user's perception of reality by superimposing computer-generated visual data onto the real world. To provide more natural interaction and spatial connection, balancing collaboration between distributed team-members closer to face-to-face collaboration, the supporting user interface should encourage remote experts to be active and on-site collaborators to be aware of their counterpart presence [2]. This can be done in several ways, such as showing the expert's messages in a shared view using visual communication cues like hand gestures, pointing, pre-defined shapes, drawing or others. Another common practice is to provide a representation, either through live stream or a virtual replica, e.g., a human avatar [1].

Nevertheless, these stimuli can lead to fractured learning experiences, in particular for guidance tasks where the on-site collaborators need to split attention between the task, the shared information, and the remote expert representation. Equally important, these cues being used at once may be considered heavy for existing networks, resulting in latency issues and communication shortage, significantly influencing attention allocation, information understanding, sense of presence and perception. Thus, the question arises: *does the expert representation really matters during scenarios of remote guidance?*

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2 USER STUDY ON SOCIAL PRESENCE

A user study was conducted to compare the level of social presence associated with the remote expert representation while team-members used two conditions (see Figure 1): C1 - Video chat and C2 - AR-based annotations. These conditions were selected due to an ongoing research collaboration with partners from the industry sector, following the insights of a participatory design process [4].

Condition C1 uses video chat features. On-site participants can point a Handheld Device (HHD) to the task context, which is shared through live video stream with the remote expert. The face of the expert is visible at all times, while the on-site participant may vary between showing the task context or their own face (back/front cameras of the device). Team-members may communicate and discuss the task context to express the main difficulties, identify areas of interest or inform where to act and what to do.

Condition C2 uses AR-based annotations as additional layers of information. On-site participants may point a HHD to capture the situation context. Using audio communication and annotation features (drawing, pre-defined shapes, notes, etc.) they can present ideas, questions or indicate intentions. The annotations are sent to the remote expert for suggesting instructions using similar features. After receiving feedback, the HHD can be placed on surface to follow the instructions in a hands-free setting. At any time, it can be picked up and the annotations viewed on top of the real world. Team-members are never displayed to one another. Yet, pop-up notifications are shown each time a collaborator is creating content, providing a sense of what is happening [5].

Both conditions were developed using the Unity 3D game engine, based on C# scripts. Communication was provided over Wi-Fi through WebRTC calls to a dedicated server. To place the augmented content in the real-world environment, we used the Vuforia library.

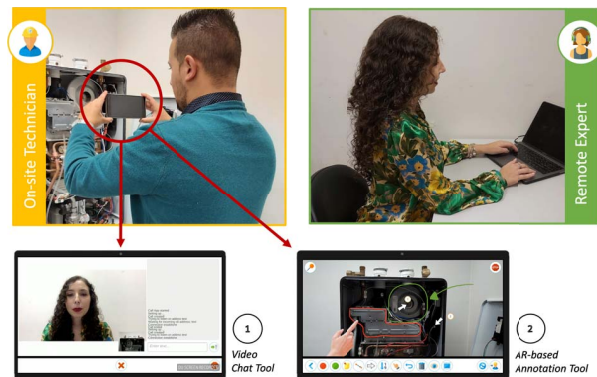


Figure 1: Remote guidance tools: 1- Video Chat; 2- AR annotations.

A within-group experimental design was used. As case study, we focused on a synchronous well defined-problem, a maintenance task with 4 completion stages: 1- replace interconnected components, 2- plug and unplug some energy modules, 3- remove a specific sensor, 4- integrate new components into the equipment. For each condition,

different alternatives were used to minimize bias, i.e., learning effect, while ensuring a similar level of difficulty and resources. Participants needed to use different hand tools to perform the procedures, although little physical movement was required.

Participants were instructed on the setup, the task, and gave their informed consent. They were introduced to both conditions and a time for adaptation was provided. Participants would act as on-site technicians, while a researcher was the remote counterpart to ensure the instructions were correctly transmitted. At the end, they answered a post-task questionnaire and a small interview was conducted to discuss their preferences and opinions. We registered the following aspects of collaboration (Likert-type scale: 1- Low; 7- High): easy to communicate or express ideas, level of social presence, enjoyment, mental effort, information understanding and attention allocation.

3 RESULTS AND DISCUSSION

We recruited 26 participants (9 female - 34.7%), whose ages ranged from 20 to 63 years old ($M = 33.1$, $SD = 11.7$). From these, 14 participants had prior experience with AR and 24 with collaborative tools. Participants had various professions, e.g., MSc and PhD students, Researchers and Faculty members from different fields. Also, Software Engineers, Front End Developers and an Assembly Line Operator. Regarding task completion time, it lasted 16 minutes on average ($SD = 2.68$) using condition C1 and 12 minutes on average ($SD = 2.66$) using condition C2.

For the majority of aspects of collaboration, i.e., easy to share ideas properly, level of attention allocation, level of information understanding, level of enjoyment and easy to communicate, condition C2 was rated higher by the participants (Figure 2). Despite, the level of social presence was slightly higher for condition C1. This might be associated to the live video stream, allowing to see each other at all times, which may have an impact on participants feeling of togetherness with their counterparts. In contrast, condition C2 did not allow to see a representation of the remote expert, which might have affected the level of social presence. As for mental effort, condition C1 was rated higher, which translated into some arguing regarding where and how to perform some activities during the study.

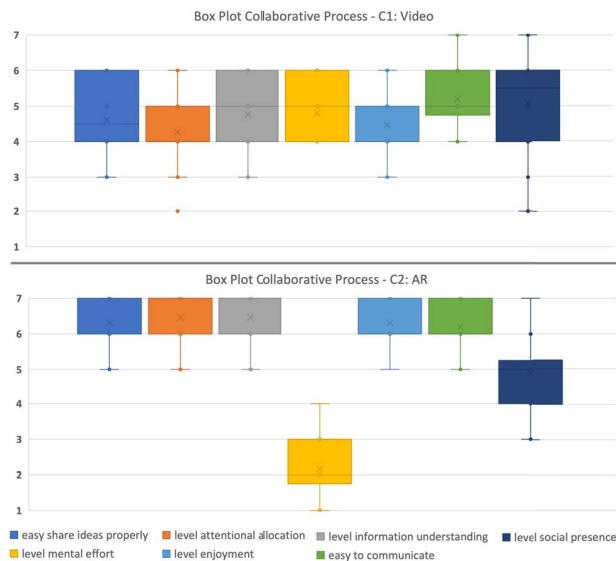


Figure 2: Collaborative process including all measures collected: easy to share ideas properly, as well as communicate, level of attentional allocation, information understanding, mental effort, enjoyment, social presence. Top - C1: Video chat; Bottom - C2: AR-based annotations. Data displayed using a Likert-type scale: 1- Low; 7- High.

The majority of participants preferred condition C2 despite not having a representation of the remote counterpart, since it enabled seeing non-verbal cues aligned with the task context, which contributed to express themselves better and having a greater perception of where to perform a given action. One participant stated that "I was expecting that the absence of video would affect my collaboration. Nevertheless, since the AR-based tool focused more on the task itself, I was engaged in such a way, that not viewing the expert did not affect me at all". Altogether, while some cases of remote collaboration supported by AR may benefit from having the expert representation available [6], our study points that for guidance scenarios like maintenance procedures, that may not be the case. Most participants considered that having the remote expert present in the user interface might not be helpful in all scenarios, leading to possible increased cognitive load, hindering the performance and work efficiency. Having only the AR-based cues and audio communication would be enough to support most tasks, while still stimulating their sense of social presence. Yet, some participants suggested viewing the remote expert, not as a basic feature, but as an option for specific use-cases could help increase empathy and trust. These results illustrate that although having a representation of remote experts may contribute to higher sense of social presence, its absence may not significantly affect the accomplishment of guidance tasks when AR annotations and notifications are used as an alternative.

4 FINAL REMARKS AND FUTURE WORK

We explored the role of social presence for remote guidance. Although the video condition presented a higher level of social presence, the remaining aspects of collaboration rated higher for the AR-based condition. In fact, this last was preferred by the majority of participants, reporting that having the remote expert visible at all times was not necessary, and could only be available as an optional feature, allowing team-members to customize their collaborative experience according to the tasks needs. This study is being expanded by planning a formal user study with domain experts, which was not possible to perform (yet) due to the pandemic constrains.

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